THE SCIENTIFIC AND TECHNICAL SUB-COMMITTEE OF THE UNITED NATIONS COMMITTEE ON THE PEACEFUL USES OF OUTER SPACE

*J.H. Carver**

I. INTRODUCTION

The Scientific and Technical Sub-Committee is one of two standing subsidiary bodies of the United Nations Committee on the Peaceful Uses of Outer Space. Established by the main Committee at its first meeting in March 1962, the Sub-Committee usually meets annually for a period of two or three weeks. Its fourteenth session was held at United Nations headquarters, New York, in February 1977. Dr. D.F. Martyn of Australia was the first chairman of the Sub-Committee and the author succeeded him in 1970. The Sub-Committee is serviced by the Outer Space Affairs Division of the United Nations Secretariat.

As with its sister body, the Legal Sub-Committee, membership of the Scientific and Technical Sub-Committee is open to all thirty-seven[1] Member States of the main Committee. With its broadly based membership, the Sub-Committee is a fairly large body which, throughout its history, has sought to conduct its work in such a way as to reach agreement by consensus without recourse to majority voting. While the search for consensus has often been slow and sometimes difficult the procedure has ensured that the Sub-Committee's recommendations have contained a realistic compromise between the views of a widely representative group of space and non-space powers.

Amongst the functions of the Sub-Committee are the exchange of information about international cooperation in outer space matters, the provision of advice to the Main Committee on scientific, technical and organizational questions and the promotion of international cooperation in outer space matters with particular reference to space applications programs which might benefit developing countries. Before examining some of the Sub-Committee's work in these fields it is necessary to review very briefly the relevant scientific and technical background to outer space activities.

---

*Elder Professor of Physics at the University of Adelaide, Australia, and Chairman of the Scientific and Technical Sub-Committee of the United Nations Committee on the Peaceful Uses of Outer Space. The views expressed in this paper are those of the author and are not necessarily connected with any organization of which he is a member.

[1]Albania, Argentina, Australia, Austria, Belgium, Brazil, Bulgaria, Canada, Chad, Chile, Czechoslovakia, Egypt, France, German Democratic Republic, Germany (Federal Republic of), Hungary, India, Indonesia, Iran, Italy, Japan, Kenya, Lebanon, Mexico, Mongolia, Morocco, Nigeria, Pakistan, Poland, Romania, Sierra Leone, Sudan, Sweden, Union of Soviet Socialist Republics, United Kingdom of Great Britain and Northern Ireland, United States of America and Venezuela.

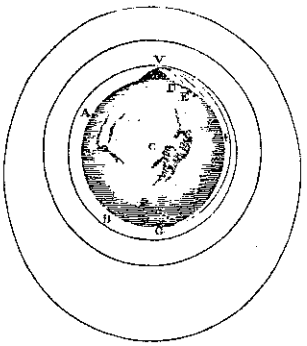
II. THE SCIENTIFIC AND TECHNICAL BACKGROUND

Sputnik I, the first artificial satellite of the Earth, was placed into orbit by the Soviet Union on October 4, 1957, and the first United States satellite was launched on January 31, 1958. In the first twenty years of the space age, men have been to the Moon; the unmanned exploration of the planets has commenced; astronomy has been freed from the limitations imposed by the Earth's atmosphere; the sun, the upper atmosphere and the outer environment of the Earth have been studied in great detail; and the application of space techniques to practical global problems has brought great improvements to meteorology, communications, navigation and Earth resources studies. That all these scientific advances and practical benefits have been achieved in so short a period is indeed impressive.

Although the space age began less than twenty years ago the scientific principles governing the motion of Earth satellites have been thoroughly understood since the time of Sir Isaac Newton (1642-1727). Some appreciation of these principles is an essential background to the work of the Scientific and Technical Sub-Committee because no amount of political dispute or legal argument can alter the natural laws of celestial mechanics which describe the motion of Earth satellites and other astronomical bodies. This is not the place to attempt to describe these scientific principles in any detail but there is, fortunately for our present purposes, a famous diagram of Newton's which illustrates most of the essential features.[2] Newton describes the motion of an object projected from a mountain which is imagined to be so high that air resistance is negligible. As the velocity of the object is increased it travels a further and further distance before falling back to the Earth under its own weight. When the velocity with which the object is projected is sufficiently great, it may "pass into space" without touching the Earth. Newton went on to point out that when the object returned to its launching point after first circumscribing the Earth it would have the same velocity as when first projected. If air resistance could be neglected the satellite would therefore "go on revolving through the heavens—just as planets do in their orbits."

---

[2] I. Newton, Mathematical Principles of Natural Philosophy and his Systems of the World 551 (A. Motte trans. 1729, F. Cajori ed. 1934). Newton's explanation of the launching of an artificial Earth satellite:



Newton's diagram illustrates one of the most fundamental properties of Earth satellites; namely, their very great speed. There is no such thing as a slowly moving Earth satellite because the velocity must exceed 8 km per second or the object will not be able to maintain a stable satellite orbit about the Earth. Satellites are intrinsically high velocity objects which may move rapidly from one region of space to another without being confined to any particular geographical region. Of course, if a satellite is placed at just the right altitude over the equator its own rapid motion may be exactly in synchronism with the rotation of the Earth so that the satellite moves in a geostationary orbit with great advantages for communication systems. Other satellite orbits may be highly eccentric, the satellite coming low into the atmosphere at perigee and sweeping afar out into space at apogee. The motion of an Earth satellite is continuous about the globe and is most compactly described not by reference to local geographical coordinates but in terms of orbital parameters which define the complete motion of the satellite around the entire Earth. These remarks are perhaps sufficient to remind us how different are the motions of artifical satellites of the Earth from the much more familiar motions of objects on the land, the sea or even in the air.

Most of the work of the Scientific and Technical Sub-Committee has been concerned with artificial satellites in orbit around the Earth rather than with more distant space probes for the very good reason that Earth orbiting satellites have already led to a wide range of practical applications which are of interest to many countries.

III. INFORMATION AND ADVICE ON INTERNATIONAL COOPERATION

One function of the Sub-Committee is to encourage the exchange of information about international cooperation in outer space matters with the object of increasing the awareness within all countries of the beneficial applications of space techniques. Continued progress in space applications depends on the nature, rate of funding and success of the space programs of those nations and groups of nations which have developed the launching capability on which all space activities, whether national or international, in the end, rely. At each of its sessions, the Sub-Committee receives reports from its members in which they review their national and cooperative international space activities during the preceding calendar year. In these national reports, and in the related general debate, States are able to explain developments in their own space programs, to state how their national programs have contributed to international activities, to comment on the effectiveness of cooperative programs in which they have participated, to suggest ways for removing difficulties that may have limited their involvement in particular projects and to express any reservations they may have concerning new applications arising from the space programs of other nations. Discussion of these topics has been wide ranging and the Sub-Committee has proved to be a successful forum used by States with both large and small space programs to express their sometimes differing views on space matters in a realistic and constructive manner.

Sessions of the Sub-Committee are attended by representatives of a large group of international bodies with interests in space research and its applications, in addition to

the delegates from Member States. For example, at the thirteenth session of the Sub-Committee held in Geneva in March-April 1976[3] the following bodies were represented: the United Nations Environment Program (UNEP), the United Nations Development Program (UNDP), the Food and Agriculture Organization of the United Nations (FAO), the United Nations Educational, Scientific and Cultural Organization (UNESCO), the World Health Organization (WHO), the International Telecommunications Union (ITU), the World Meteorological Organization (WMO), the Inter-Governmental Maritime Consultative Organization (IMCO), the European Space Agency (ESA), the Committee on Space Research (COSPAR), and the International Astronautical Federation (IAF). Reports to the Sub-Committee from these specialized bodies have made a most valuable contribution to the exchange of information about international space activities and this expert advice has greatly strengthened the Sub-Committee in its role as a "focal point" for discussion of international cooperation in the peaceful uses of outer space.

The Committee on Space Research (COSPAR) of the International Council of Scientific Unions and the International Astronautical Federation (IAF) have close links with the Scientific and Technical Sub-Committee. Both of these organizations have advised the Sub-Committee on various scientific and technical questions and have cooperated in joint projects with the United Nations including the publication of scientific data and the holding of technical panels on space related matters. The professional advice received from COSPAR and IAF has been of great value to the Sub-Committee.

The Sub-Committee reports to the main Committee which it advises on scientific, technical and organizational matters concerned with international cooperation in the peaceful uses of outer space. It is important to remember that the Sub-Committee provides its scientific and technical advice in the political context of a government to government meeting.The delegates to the Scientific and Technical Sub-Committee are instructed by the same governments that send delegates to the main Committee and to the Legal Sub-Committee; all three bodies exist to consider outer space matters which require discussion at a governmental level. Unlike, for example, COSPAR and the IAF, the Scientific and Technical Sub-Committee is not an international college holding scientist to scientist and engineer to engineer meetings. The work of the Sub-Committee does, however, benefit considerably from the traditions of scientific and technical cooperation which have been fostered in international meetings of the collegiate type. And the members of the Sub-Committee and their advisers share a common understanding of the scientific and technical principles on which the Sub-Committee's work is based.

---

[3]*See* U.N. Doc. A/AC.105/170 (1976).

IV. LAUNCHING RANGES UNDER UNITED NATIONS SPONSORSHIP

The Sub-Committee does not, of course, have the resources to function in any way as an operational space agency. Its role is to coordinate United Nations and other international space activities and to promote international cooperation in outer space matters. One of the ways in which the Sub-Committee has sought to increase international cooperation in the use of space facilities has been by recommending the granting of United Nations sponsorship to sounding rocket ranges. A Member State may request United Nations sponsorship for its range under the provisions of a resolution passed by the General Assembly in December 1962. United Nations sponsorship has been granted so far to two rocket ranges, the Thumba Equatorial Rocket Launching Station of the Vikram Sarabhai Space Centre in India and the CELPA Mar del Plata Rocket Launching Station in Argentina. The Sub-Committee annually receives reports on the use of these sounding rocket facilities for international cooperation and training in the peaceful scientific exploration of outer space.

V. PRACTICAL APPLICATIONS OF SATELLITES

The practical benefits to be obtained from communications and meteorological satellites were already established over a decade ago when such satellites began to be used to provide regular operational services in a number of countries. At the same time it was realized that satellites might also have a number of other important practical applications.

In 1967 a Navigational Satellite Working Group, formed by the Committee on the Peaceful Uses of Outer Space, reported that it would be technically feasible to develop a navigational services satellite system to meet particular needs of civil aviation and sea-borne traffic and to help resolve basic navigational requirements.[4] Following discussions in the Inter-Governmental Maritime Consultative Organization (IMCO), which were reported to the Scientific and Technical Sub-Committee, the International Maritime Satellite System (INMARSAT), has now been established to facilitate the use of satellites for maritime navigation.

In 1968 the General Assembly of the United Nations set up a Working Group on direct broadcast satellites to study the technical feasibility of communications by direct broadcasting from satellites, as well as the social, legal, cultural and other implications of such communications. The Working Group concluded, *inter alia*, that direct broadcasting into community receivers could be technically feasible in the mid-1970's. This conclusion has been borne out by the success of the satellite instructional television experiment (SITE) currently being undertaken by India using the United States ATS-6 satellite. Reports on the SITE program at the 1976 session of the Scientific and Technical Sub-Committee indicated how isolated village communities could benefit from direct

---

[4]*See* U.N. Doc. A/AC.105/38 (1967).

satellite broadcasting designed to promote improvements in agriculture, health and cultural understanding.[5]

Matters relating to the practical applications of space techniques, particularly the new applications to remote sensing, have now become of major concern to the Scientific and Technical Sub-Committee.

VI. UNITED NATIONS SPACE APPLICATIONS PROGRAM

The first United Nations Space Conference which was held in Vienna in August 1968[6] focused the attention of many countries on the potential benefits which might be obtained from the practical applications of space techniques. Largely as a result of this Conference, the Scientific and Technical Sub-Committee actively sought to encourage international cooperation in the field of space applications with particular reference to programs which would enable the developing countries to share in the advantages to be gained from this new technology.

In order to achieve this objective, the Sub-Committee recommended that the Secretary-General appoint to the Outer Space Affairs Division a qualified individual whose full-time task it would be to promote the practical applications of space technology. The main Committee on the Peaceful Uses of Outer Space and the General Assembly accepted the recommendation and in March 1970 the Secretary-General appointed the first United Nations Expert on Space Applications. It is fortunate indeed that it has been possible to attract to this demanding post persons of the quality of Professor H. Ricciardi of Argentina and his successor Professor H. Murthy of India.

The tasks which the Sub-Committee set for the Expert on Space Applications were formidable. In order to promote space applications particularly amongst developing countries, the Expert was required to liaise with all components of the United Nations family and maintain contact with all Member States seeking information and assistance relevant to United Nations programs in this field. In reporting to the Sub-Committee, one of the Expert's first recommendations was the adoption of a United Nations Space Applications Program designed to create awareness on the part of policy makers and interested government agencies of benefits from space applications technology and to encourage training and education programs which would enable officials from developing countries to gain experience in the practical applications of space technology.

This Space Applications Program has been supported by the Sub-Committee. Initially the program concentrated on creating a broad awareness of the potential benefits of space applications but recent work has been increasingly directed towards practical training courses rather than more generally oriented seminars. In response to a

---

[5]*See* U.N. Doc. A/AC.105/170 (1976).

[6]Proceedings reported in 1, 2 A/CONF. 34/2 (1968).

questionnaire Member States have indicated a particular need for education and training in applications concerned with satellite communications and broadcasting, remote sensing and meteorology.

In implementing the Space Applications Program the Expert has had extensive discussions with officials in many countries about the various possibilities for development that are created by space technology. An important point of his work has been to encourage the establishment of technical panels composed particularly of representatives from developing countries to observe and study various space applications projects. The Expert maintains close contacts with the specialized agencies and with international governmental and nongovernmental organizations concerned with international cooperation in the peaceful uses of outer space. Consultations are arranged by the Expert to secure sponsorship of panels by these agencies and organizations and to seek Member States to host the meetings.

In response to offers extended by Member States, panels, seminars or workshops were organized during 1971-75 in Argentina, Brazil, Canada, Egypt, Ethiopia, France, India, Indonesia, Italy, Japan, Kenya, Mexico, and the United States.[7] Further offers were made for the period 1976-77 for panels, seminars or workshops to be convened in the Federal Republic of Germany, Pakistan, the United Kingdom and the Union of Soviet Socialist Republics.

Following a request made by the Sub-Committee at its 1975 session, the United Nations, in cooperation with FAO, arranged for the training during 1976 of specialists from developing countries in the area of remote sensing including the interpretation of US Landsat images, aircraft-derived data and ground truth information with particular reference to the solution of problems related to the participant's own geographical region.

---

[7]United Nations Panel Meeting on Remote Sensing Systems for Earth Resources Survey (Ann Arbor, United States, May 3-14, 1971); United Nations Panel Meeting on the Establishment and Implementation of Research Programs in Remote Sensing (Sao Jose dos Campos, Brazil, November 29-December 10, 1971); United Nations/WMO Panel and Training Seminar on Uses of Meteorological Satellite data - with special emphasis on tropical areas (Mexico City, November 29-December 8, 1972); Regional United Nations Panel Meeting on Satellite Instructional Television systems (New Delhi and Ahmedabad, India, December 12-20, 1972); Summer School on Remote Sensing of Earth Resources (Tarbes, France, August 21-September 21, 1973); United Nations/UNESCO Regional Seminar on Satellite Broadcasting Systems for Education and Development (Addis Ababa, Ethiopia, October 21-31, 1973); United Nations Panel Meeting on the Applications of Remote Sensing of Earth Resources (Buenos Aires, Argentina, December 2-8, 1973); United Nations Panel Meeting on Satellite Broadcasting Systems for Education (Tokyo, Japan, February 26-March 7, 1974); United Nations/FAO Regional Seminar on Remote Sensing of Earth Resources and Environment (Cairo, Egypt, September 4-13, 1974); United Nations Inter-regional seminar on the Applications of Geodetic and Remote Sensing Data from Satellite for Cartography - surveying and mapping (Sao Jose dos Campos, November 4-15 1974); United Nations Inter-regional Technical Seminar on Remote Sensing Applications in cooperation with Canada and UNESCO (Guelph, Canada, May 12-30, 1975); Joint United Nations/UNESCO Regional Seminar on Satellite Broadcasting Systems for Education and Development (Mexico City, September 2-11, 1975); Joint United Nations/WMO Regional Training Seminar on the Interpretation, Analysis and Use of Meteorological Satellite Data (Nairobi, Kenya, October 6-17, 1975); United Nations/FAO Regional Seminar on Remote Sensing (Jakarta, Indonesia, November 19-28, 1975).

The Sub-Committee annually reviews the progress of the United Nations Programme of Space Applications and considers proposals made by the Expert for future activities. In support of education and training through the Space Applications Programme, the Sub-Committee has been most appreciative of the offers of fellowships made by various Member States including Austria, Belgium, Brazil, India, Italy, Japan, the United Kingdom and the United States.

VII. COORDINATION OF THE UNITED NATIONS SPACE ACTIVITIES

As evidenced by the large group of international bodies which attend sessions of the Scientific and Technical Sub-Committee, involvement in space activities is spread widely throughout the United Nations system. Mindful of its own role as a "focal point" for international cooperation in outer space matters the Sub-Committee has drawn attention to the need to ensure effective coordination of the activities of the United Nations and the specialized agencies in the area of space applications.

In 1974 the Sub-Committee requested the Secretary-General to report and comment on the whole question of the coordination of space application activities within the United Nations system. At its 1976 session the Sub-Committee welcomed the new administrative machinery reported by the Secretary-General to improve interagency coordination in outer space matters. The Sub-Committee indicated that it wished to continue to be kept informed of the steps being taken to coordinate the outer space activities of organizations within the United Nations system.

VIII. REMOTE SENSING OF THE EARTH BY SATELLITE

At its recent sessions the Scientific and Technical Sub-Committee has given considerable attention to questions relating to the remote sensing of the Earth by satellite. Remote sensing could bring significant benefits to many countries, particularly those in developing areas, provided there is a sufficiently widespread understanding of the potential benefit of this space technique. Global monitoring using remote sensing techniques could also be of great assistance in the preservation of the Earth's environment.

Recognizing the importance of remote sensing, the General Assembly in November 1971 endorsed the convening by the Scientific and Technical Sub-Committee of a Working Group on remote sensing of the Earth by satellites. The Working Group, under the vigorous chairmanship of the late Franco Fiorio of Italy, studied various means of furthering international cooperation in this field, summarized scientific and technical developments concerning remote sensing systems and reviewed methods for the collection, processing and interpretation of data.[8] A Task Force, open to

---

[8] *See* U.N. Doc. A/AC.105/102 (1972).

all members of the Working Group, studied the documentation submitted for the consideration of the Working Group, analyzed responses to a questionnnaire on remote sensing sent to Member States and reviewed proposed alternatives for the dissemination and optimum utilization of remotely sensed data.

The possibility of establishing under United Nations auspices an international global centre for remote sensing, as well as setting up regional data acquisition and/or data storage and dissemination centers, with the possible attachment of education and training facilities to such centers, were important issues raised in the Working Group report. The Sub-Committee, realizing the complexity of these matters, requested that the Secretary-General undertake studies concerning their organizational and financial implications.

Reports on these studies were before the Sub-Committee at its thirteenth session in 1976. The Sub-Committee affirmed the value of keeping in mind the distinction between the current *pre-operational*/experimental phase of remote sensing and a possible future global/international *operational* remote sensing system or systems. Technical, organizational and financial aspects are likely to be major distinguishing factors between these phases.

The present pre-operational experience has arisen largely in connection with the United States Landsat system. While it is possible that other systems may be set up in the next few years, these, too, would be of an experimental nature and fully operational systems are unlikely until the 1980's. The Sub-Committee has received reports from a number of users on the experience which they have gained in the present pre-operational phase. Discussions within the Sub-Committee based on this growing experience should assist in defining the parameters of a possible operational phase which will best satisfy the interests of most users.

No commitment has yet been made to establish an operational system of remote sensing of the Earth by satellite. Discussions within the Sub-Committee have suggested, however, that satellite remote sensing systems will in time become, like weather and communications satellite systems, an integral part of the planning and production of national economies and that international cooperation will be essential at that stage as the only cost-effective approach for the majority of countries. One of the major tasks facing the Sub-Committee in the immediate future is to ensure that the means are provided to allow this international cooperation to be achieved. This will require detailed investigation of the possible forms for the space and ground segments of a future operational phase of satellite remote sensing of the Earth.

The Sub-Committee has begun to outline some of the possible operational alternatives and is seeking to clarify such basic questions as whether or not a future remote sensing operational space segment would encompass one or more nationally owned and operated systems; and to what extent the United Nations or other international organization(s) should have a coordinating role. The Sub-Committee

considers that ground facilities for receiving, pre-processing and disseminating data in the operational phase would be, in all probability, financed, owned and operated by individual users and associations of users. It noted that in the case of existing satellites global coverage could be achieved with a limited number of ground stations. For example, approximately 15 ground stations, properly distributed, would be sufficient to provide complete global coverage in the case of the US Landsat system. Questions of economy and the ready availability of data to local users will be amongst the many factors that need to be considered in planning future operational systems.

Future operational systems for satellite remote sensing of the Earth will need to take account of the interests of a very wide range of users in many countries at different stages of development. In the report of its thirteenth session, the Sub-Committee pointed out:

> that users' needs and interests would have a decisive influence in the future realization of operational systems for remote sensing from satellites. Thus, technology which already could be seen to provide a great potential for social and economic development of both developing and developed countries, was likely to bring significant benefits of an interdisciplinary nature in such major fields as: The mapping of areas of the world, and the recording of changes in the conditions and use of the earth's surface; Agricultural forecasting as an aid to crop, forest and animal production, and agricultural supplies and products distribution; Geological mapping to facilitate the efficient exploration and development of mineral resources; Hydrological surveys for water resources identification, to permit urban, rural and agricultural image planning and pollution monitorings; Land uses surveys to permit efficient development of urban and rural areas, and to aid in transmigration planning.[9]

Reiterating the view that further study of organizational, technical and financial matters should progress together with consideration of the legal aspects of remote sensing, the Scientific and Technical Sub-Committee asked the Secretariat to prepare further studies on remote sensing questions for consideration at the 1977 session of the Sub-Committee.

IX. FUTURE WORK

At each of its recent sessions the Scientific and Technical Sub-Committee has carefully reviewed its future work and delineated certain items to which it planned to give special attention at the following session. This has been a healthy exercise which has focused the Sub-Committee's attention on particular items of current concern.

In view of the complexity of the issues that have already been raised, it seems most likely that remote sensing of the Earth by satellites will be of major concern to the Sub-Committee for some time to come. The Sub-Committee also has a continuing interest in the Space Applications Programme and educational and training matters related to it.

---

[9]U.N. Doc. A/AC.105/170 (1976).

The Sub-Committee has begun to consider the options relating to a possible second United Nations Conference on Outer Space matters and these options will be discussed further at the 1977 session.

These are all matters which are on the current agenda of the Scientific and Technical Sub-Committee. In the longer term it seems probable that the Sub-Committee will continue to concentrate most of its attention on the practical applications of space technology because of the exciting possibilities for developments that are inherent in this new technology. The value of the applications of space techniques to meteorology, communication, direct broadcasting, navigation and Earth resources is now widely recognized and many countries derive substantial benefits from space technology.

The Sub-Committee will need to continue its efforts to see that the benefits of these well-recognized space applications are even more widely shared amongst the countries of the world. Many other beneficial applications of the space technique are of course possible. Materials processing in space and the utilization of solar energy are examples of applications which could be of profound importance if there are favorable technological developments. In planning its future work the Scientific and Technical Sub-Committee will need to remain alert to applicatons which may follow further innovations in space techniques.